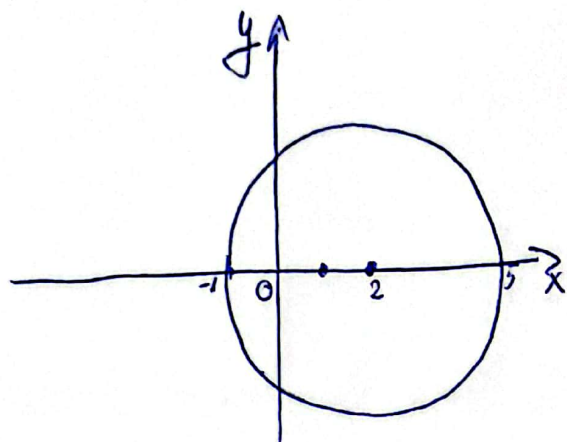


Dim ① $\hat{z}$ ② $\Rightarrow J = J_1 + J_2 = \frac{16\pi i}{125} - \frac{16\pi i}{125} = 0$

6) a) $\int_{|z-2|=3} \frac{z-2}{z^2+2z-3} dz$

$|z-2|=3 \Rightarrow z_0=2, r=3$



$z^2 + 2z - 3 = 0 \Leftrightarrow z(z-1) + 3(z-1) = 0 \Leftrightarrow (z+3)(z-1) = 0$

$\Rightarrow \begin{cases} z_1 = -3 \Rightarrow A(-3, 0) \notin D \\ z_2 = 1 \Rightarrow B(1, 0) \in D \end{cases}$

$\Rightarrow B(1, 0) \in D$ pol de ordin $\underline{1}$

Dim th. reziduurilor:

$\int_{\gamma} z^{-2} \pi i \operatorname{res} \left\{ f, -3 \right\}$

$\int_{\gamma} z^{-2} \pi i \operatorname{res} \left\{ f, 1 \right\}$

$z=1$ - pol. de ord. I

$$\operatorname{res}\left\{ f, 1 \right\} = \left. \frac{z+1-2}{(z^2+z-3)'} \right|_{z=1} = \frac{z+1-2}{2z+2} \Big|_{z=1}$$

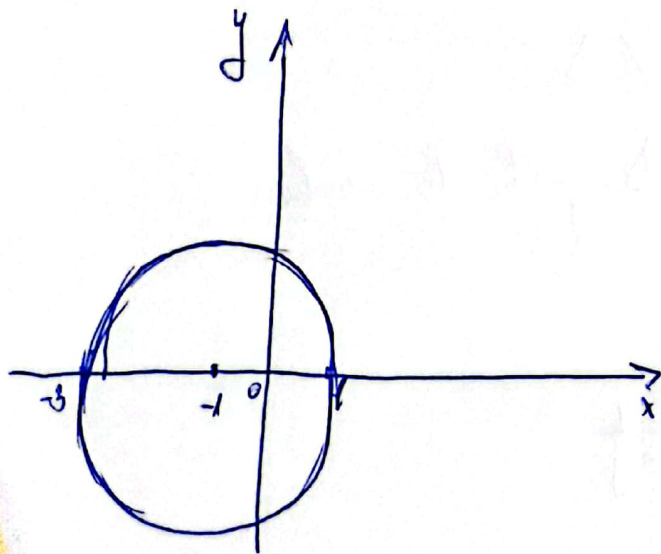
$$= \frac{1+1-2}{2+2} = \frac{-1}{4}$$

$$J = 2\pi i \cdot \left(\frac{-1}{4} \right) = -\frac{\pi i}{2}$$

$$b) \int_{|z+1|=2} \frac{z+2}{z^2(z+1)(z-3)} dz$$

$$|z^2+1|=2 \Rightarrow z_0 z = -1$$

$$R=2$$



$$z^2(z+1)(z-3)=0 \Rightarrow \begin{cases} z_1 = z_2 = 0 \Rightarrow A(0,0) \text{ pol de ord } \underline{2} \\ z_3 = -1 \Rightarrow B(-1,0) \text{ pol de ord } \underline{1} \\ z_4 = 3 \Rightarrow C(3,0) \notin D \end{cases}$$

$$J = 2\pi i \left[\text{res} \left\{ f, 0 \right\} + \text{res} \left\{ f, -1 \right\} \right]$$

$$z_1, z_2 = 0 \rightarrow \text{pol de ord. II}$$

$$\text{res} \left\{ f, 0 \right\} = \frac{1}{1!} \cdot \lim_{z \rightarrow 0} \left[(z-0)^2 \cdot \frac{z+2}{z^2(z+1)(z-3)} \right]$$

$$= \lim_{z \rightarrow 0} \left[\frac{z+2}{(z+1)(z-3)} \right]$$

$$= \lim_{z \rightarrow 0} \frac{-z^2 - 4z + 1}{(z^2 - 2z - 3)^2} = \frac{1}{(-3)^2} = \frac{1}{9}$$

$$z_3 = -1 \rightarrow \text{pol de ord. I}$$

$$\text{res} \left\{ f, -1 \right\} = \frac{z+2}{(z^2(z+1)(z-3))'} \Big|_{z=-1}$$

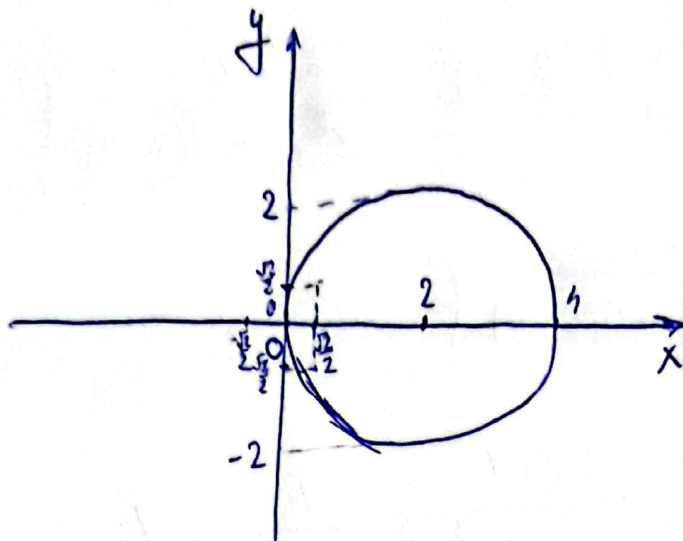
$$= \frac{z+2}{4z^3 - 6z^2 - 6z} \Big|_{z=-1}$$

$$= \frac{1}{-4 - 6 + 6} = -\frac{1}{4}$$

$$J = 2\pi i \left(\frac{1}{9} - \frac{1}{4} \right) = 2\pi i \cdot \frac{-5}{36} = -\frac{5\pi i}{18}$$

$$c) \int_{|z-2|=2} \frac{1}{z^h+1} dz$$

$$|z-2|=2 \Rightarrow z_0=2, h=2$$



$$z^4+1=0 \Rightarrow z^4=-1$$

$$z_1=-1 \Rightarrow \begin{cases} a=-1 \\ b=0 \end{cases} \Rightarrow r=1$$

$$\begin{cases} \cos \alpha = \frac{a}{r} = -1 \\ \sin \alpha = \frac{b}{r} = 0 \end{cases} \Rightarrow \alpha = \pi$$

$$z_1 = \cos \pi + i \sin \pi$$

$$z_k = \cos\left(\frac{\pi + 2k\pi}{4}\right) + i \sin\left(\frac{\pi + 2k\pi}{4}\right)$$

$$, k \in \{0, 1, 2, 3\}$$

$$z_0 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \in D \rightarrow \text{pol de ord. } 1$$

$$z_1 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \notin D$$

$$z_2 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i \notin D$$

$$z_3 = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i \in D \rightarrow \text{pol de ord. } 1$$

$$\gamma = \text{res} \left[f, \frac{\sqrt{2}}{2}(1+i) \right] + \text{res} \left[f, \frac{\sqrt{2}}{2}(1-i) \right]$$

$$\text{res} \left[f, \frac{\sqrt{2}}{2}(1+i) \right] = \frac{1}{(z^4+1)'} \Big|_{z=\frac{\sqrt{2}}{2}(1+i)} = \frac{1}{4z^3} \Big|_{z=\frac{\sqrt{2}}{2}(1+i)}$$

$$= \frac{1}{4 \left(\frac{\sqrt{2}}{2} (1+2i+\sqrt{2}) \right)} = \frac{1}{4 \cdot \frac{1}{\sqrt{2}} \cdot 2i} = \frac{1}{4i}$$

$$= \frac{1}{\cancel{4} \cdot \frac{2\sqrt{2}}{8} \cdot (1+i)^3} = \frac{1}{\sqrt{2}(1+i)^3}$$

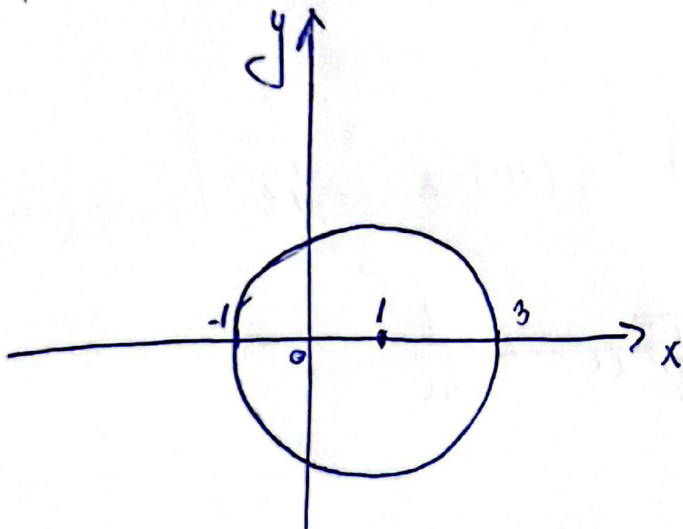
$$\text{res} \left[f, \frac{\sqrt{2}}{2}(1-i) \right] = \frac{1}{(z^4+1)'} \Big|_{z=\frac{\sqrt{2}}{2}(1-i)} = \frac{1}{\cancel{4} \cdot \frac{2\sqrt{2}}{8} (1-i)^3}$$

$$= \frac{1}{\sqrt{2}(1-i)^3}$$

$$\begin{aligned}
 J &= 2\pi i \left(\frac{1}{\sqrt{2}(1+i)^3} + \frac{1}{\sqrt{2}(1-i)^3} \right) = \frac{2\pi i}{\sqrt{2}} \left(\frac{1}{-2+2i} + \frac{1}{-2-2i} \right) \\
 &= \frac{\pi i}{\sqrt{2}} \left(\frac{1}{-1+i} + \frac{1}{-1-i} \right) = \frac{\pi i}{\sqrt{2}} \frac{1+i + 1-i}{(1+i)(-1-i)} = \frac{\sqrt{2}}{2} \cdot \pi i \cdot \frac{2}{-1-i-1-i} \\
 &= -\frac{\sqrt{2}}{2} \pi i
 \end{aligned}$$

$$d) \int_{|z-1|=2} \frac{z+3}{(z^2-6z+8)^2} dz$$

$$|z-1|=2 \Rightarrow z_0=1, \quad R=2$$



$$(z^2-6z+8)^2=0 \Rightarrow z^2-6z+8=0$$

$$\Delta = 36 - 32 = 4$$

$$\begin{aligned}
 z_{1,2} &= \frac{6 \pm 2}{2} \Rightarrow \begin{cases} z_{1,2} = \frac{8}{2} = 4 \notin \Delta \\ z_3 = z_4 = \frac{4}{2} = 2 \in \Delta \end{cases} \Rightarrow \text{pol de ord } 2
 \end{aligned}$$

$$J = 2\pi i \operatorname{res}\{f, 2\}$$

$$\operatorname{res}\{f, 2\} = \frac{1}{1!} \lim_{z \rightarrow 2} \left[(z-2)^2 \cdot \frac{z+3}{(z-2)^2(z-4)^2} \right]'$$

$$= \lim_{z \rightarrow 2} \left(\frac{z+3}{z^2-8z+16} \right)'$$

$$= \lim_{z \rightarrow 2} \frac{z^2-8z+16 - (z+3)(2z-8)}{(z^2-8z+16)^2}$$

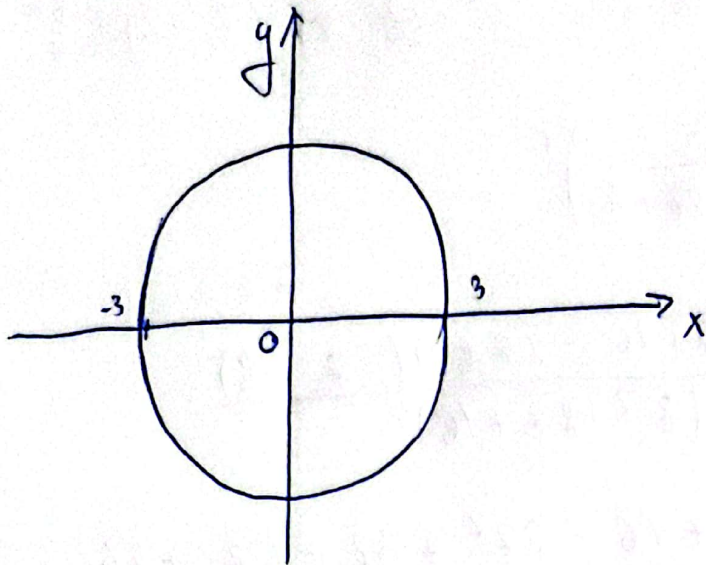
$$= \lim_{z \rightarrow 2} \frac{z^2 - 8z + 16 - 2z^2 + 8z - 6z + 24}{(z^2-8z+16)^2}$$

$$= \lim_{z \rightarrow 2} \frac{-z^2 - 6z + 40}{(z^2-8z+16)^2} = \frac{24}{16} = \frac{3}{2}$$

$$\Rightarrow J = 2\pi i \cdot \frac{3}{2} = 3\pi i$$

$$e) \int_{|z|=3} \frac{1}{(z^2-4)(z^2+4)} dz$$

$$|z|=3 \Rightarrow z_0 = 0, R=3$$



$$(z^2-4)(z^2+4) = 0 \Rightarrow \begin{cases} z^2-4=0 \\ z^2+4=0 \end{cases}$$

$$z^2-4 \Rightarrow \begin{cases} z_1 = 2 \in D \rightarrow \text{pol de ord } 1 \\ z_2 = -2 \in D \rightarrow \text{pol de ord } 1 \end{cases}$$

$$z^2+4=0 \Rightarrow z^2=-4 \Rightarrow \begin{cases} z_3 = 2i \in D \rightarrow \text{pol de ord } 1 \\ z_4 = -2i \in D \rightarrow \text{pol de ord } 1 \end{cases}$$

$$J = 2\pi i \left[\text{res}(f, 2) + \text{res}(f, -2) + \text{res}(f, 2i) + \text{res}(f, -2i) \right]$$

$$\operatorname{Res}\left(\int, 2\right) = \frac{z^2 + 4}{(z^2 - 4)'} \Big|_{z=2} = \frac{z^2 + 4}{2z} \Big|_{z=2} = \frac{4+4}{4} = 2$$

$$\operatorname{Res}\left(\int, -2\right) = \frac{z^2 + 4}{(z^2 - 4)'} \Big|_{z=-2} = \frac{z^2 + 4}{2z} \Big|_{z=-2} = \frac{4+4}{-4} = -2$$

$$\operatorname{Res}\left(\int, 2i\right) = \frac{z^2 - 4}{(z^2 + 4)'} \Big|_{z=2i} = \frac{z^2 - 4}{2z} \Big|_{z=2i} = \frac{-4-4}{4i} = -\frac{2}{i} = 2i$$

$$\operatorname{Res}\left(\int, -2i\right) = \frac{z^2 - 4}{(z^2 + 4)'} \Big|_{z=-2i} = \frac{z^2 - 4}{2z} \Big|_{z=-2i} = \frac{-4-4}{-4i} = \frac{2}{i} = -2i$$

$$2\pi \int = 2\pi i (2 - 2 + 2i - 2i) = 0$$